

AI-EMS User Manual

Installation, role-based operation, and demonstration guidance for AI-EMS v1.7.0

Repository: <https://github.com/Sparkling-Flames/AI-EMS>

Formal User Manual

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1. Introduction

AI-EMS is a Uniapp and UniCloud proof-of-concept for a university educational management system. It is intended for controlled demonstration and course project evaluation, not for production academic administration.

The system connects role-based dashboards, leave requests, attendance synchronization, anonymous course evaluation, course materials, profile-change review, course publishing, timetable review, and AI assistant support.

Primary users

- Students use the system to review academic progress, submit leave requests, read materials, evaluate completed courses, update editable profile fields, and ask assistant questions.
- Teachers use the system to review assigned courses, maintain attendance, approve or reject leave requests, view anonymous evaluation summaries, and manage course materials.
- Administrators use the system to create accounts, publish courses, review profile changes, and inspect system summaries.

Current scope

- The database is UniCloud NoSQL. MySQL is not used in this PoC.
- The login flow is project-specific and does not implement production SSO or full uni-id.
- The assistant can use local system context and knowledge-base retrieval. DeepSeek or OpenAI-compatible access is optional when an API key is configured. The AI assistant should be described as a support assistant, not as an official academic decision-maker.
- The system does not generate official academic documents.
- The leave workflow supports student leave submission and teacher or administrator review. When a leave request is approved, the related attendance record can be synchronized to `on\_leave`. Leave cancellation and deletion are supported where the current workflow allows them.
- Attendance support is currently focused on teacher/admin attendance maintenance and leave-to-attendance synchronization. Location-based student self check-in should not be presented as a completed end-user workflow in the current manual.
- Course evaluation supports multi-dimensional student feedback and anonymous summary views for teachers and administrators. The current system demonstrates privacy-aware evaluation display, but it is not a formal privacy-compliance solution.
- Course materials support teacher or administrator creation and update of material records, visibility control for students, and student access to visible materials and course timelines. File availability depends on the UniCloud storage/runtime environment.
- Administrator management currently covers account creation and update, account deletion where allowed, course publishing, course update/deletion, teacher assignment, schedule slots, classroom selection, cohort filtering, and timetable inspection.
- Profile-change review is supported for editable student or teacher profile fields. Users submit change requests, and administrators review them before approved changes are applied.

- The project does not currently provide official academic document generation, full production RAG with vector databases, payment processing, dormitory management, campus security, biometric attendance hardware, or parent-portal functions.

2. Getting Started

Prerequisites

Requirement	Purpose
HBuilderX	Open the Uniapp project, run H5 preview, and upload UniCloud functions.
Aliyun UniCloud service space	Run cloud functions, database collections, and cloud storage workflows.
Modern browser	Open and test the H5 preview.
Node.js and npm	Run optional local preview, lint, and tests.
Git	Clone or update the project repository.

Installation

1. Clone or download the repository from GitHub.
2. Open HBuilderX and choose File > Open Directory.
3. Select the AI-EMS project root.
4. Confirm that src, common, components, static, and uniCloud-aliyun are present.
5. If command-line preview or tests are needed, run npm install from the project root.

Running the program

The recommended run path is HBuilderX H5 preview. The optional command-line preview can be started with npm run dev:h5:local after dependencies are installed.

Login and role routing

The login page calls auth-login. After successful authentication, the session is stored locally and the user is routed by role.

Role	Landing page
student	/pages/student/dashboard

teacher	/pages/teacher/dashboard
admin or academic_staff	/pages/admin/dashboard

Provided administrator account: admin001 / AiEms2026!. Additional student and teacher accounts can be created by an administrator or imported into UniCloud with matching profile records.

3. Student User Guide

Student dashboard

The student dashboard is the main student workspace. It shows academic progress, attendance, courses, materials, recommendations, profile information, and workflow status.

- Review GPA, credits, required modules, attendance rate, alerts, and percentile information.
- Review profile fields such as student number, major, class, email, phone, and address.
- Submit editable profile changes for administrator review instead of directly changing protected records.
- Choose a teacher for a multi-teacher course offering when the course requires teacher selection.

Submit a leave request

1. Open Leave Workflow.
2. Select the related course.
3. Choose the leave type, date, and reason detail.
4. Submit the request and wait for teacher or administrator review.
5. If the request is approved, the related attendance record is synchronized to on_leave.
6. Cancel a pending or approved request when cancellation is allowed.

Course evaluation

Students can evaluate completed courses anonymously after the course has ended in the current data. When a course is associated with more than one teacher, the student should select the relevant teacher so that the feedback is counted under the correct teaching group.

- Select a completed course.
- Choose the relevant teacher when the course has multiple teachers.
- Enter six scores: content, teaching method, difficulty, workload, achievement, and overall.
- Write feedback text and submit.
- Students, teachers, and administrators can view aggregate results and anonymous comments, but individual student identities are not displayed.

Course materials

Students can open the Materials page to view visible course resources and the session timeline. Material downloads depend on the runtime and cloud storage availability.

AI assistant

Students can use the AI Assistant to ask about courses, attendance, leave status, academic progress, visible materials, and available knowledge-base content. The assistant may use configured AI model access together with retrieved system context, or fall back to local knowledge when external AI access is unavailable. It provides guidance only and does not create, approve, or modify official academic records.

4. Teacher User Guide

Teacher dashboard

The teacher dashboard focuses on assigned courses, attendance editing, leave requests, evaluation summaries, course materials, profile-change workflow, and AI assistant access.

Attendance editor

1. Select an assigned course.
2. Select a class session.
3. Review enrolled students.
4. Set editable statuses: Present, Late, Absent, or Excused.
5. Leave records with status on_leave are preserved and cannot be manually overwritten.
6. Click Save Attendance.

Leave review

Teachers can approve or reject pending leave requests for courses they teach. The optional comment field can record a short explanation. If a leave request is approved, the system updates the related attendance record to `on\_leave`. If the request is rejected, the attendance record remains unchanged.

Evaluation summary

Teachers can review aggregated course evaluation summaries and anonymous feedback for courses they are allowed to access. Student identity is not shown.

Course materials

Teachers can create or update course material metadata when they have permission for the course. Materials can be marked visible to students. If a course has a material upload deadline(now you can view as the deadline of the course), teachers cannot upload after the deadline.

AI assistant

Teachers can use the AI Assistant to ask questions about assigned courses, student rosters, attendance status, leave requests, course materials, evaluation summaries, available knowledge-base content or the policies that school have released. When an API key is configured, the assistant can call the selected chat model with retrieved project context. Without an API key, it provides a controlled local answer or fallback response based on available system data and knowledge-base records.

The assistant is for teaching support and information lookup only. It does not approve leave requests, change attendance records, edit course materials, modify evaluation summaries, or make official academic decisions.

5. Administrator User Guide

Admin dashboard

The administrator dashboard provides system metrics, leave queues, profile-change reviews, course quality monitoring, and navigation to management tools.

Account management

1. Open Admin Management.
2. Enter username, initial password, role, display name, email, and phone.
3. For students, add student number, major, class, and enrollment year.
4. For teachers, add teacher number, department, title, and office.
5. Submit the form to create the user and linked profile together.

Administrators can reset passwords from the Recent Accounts action. The current admin account cannot delete itself.

Course publishing

A course offering requires course code, course name, credits, capacity, target major, cohort year, date range, total sessions, schedule slots, classrooms, and assigned teachers.

The system validates date range, total session count, schedule slot overlap within the same course, same-major and same-cohort time conflicts, classroom conflicts, and capacity. The current implementation does not validate teacher time availability across different courses.

When publishing a course, administrators enter the course code, course name, credits, capacity, target major, cohort year, course date range, total number of sessions, weekly schedule slots, classrooms, and assigned teachers.

Before saving the course, the system checks that the date range can generate enough class sessions, that schedule slots do not overlap, that the same major and cohort do not have conflicting classes, that classrooms are not double-booked, and that the course capacity is valid. In the current implementation, the system does not check whether an assigned teacher is already teaching another course at the same time.

Cohort timetable

The timetable view lets administrators review scheduled sessions for a selected major and cohort year. It is mainly used to confirm the course schedule after publishing, while conflict checks are handled during the course-saving process.

Profile-change review

Students and teachers submit editable profile fields for review. Administrators approve valid changes or reject invalid changes with a review comment. Approved changes are applied to the linked students or teachers record.

Course and account deletion

Course deletion removes related sessions, enrollments, attendance, leave requests, evaluations, and materials. Account deletion removes linked profile and related role data according to the current implementation.

6. AI Assistant Usage

The assistant combines local knowledge-base retrieval, role-accessible system context, and optional external model access. API settings are stored locally in the browser and sent with each request when configured.

Without an API key

- The assistant searches local knowledge_base records and safe system context.
- If a reliable local answer exists, it returns that answer with a source title where available.
- If no reliable answer exists, it returns a fallback asking the user to configure an API key or contact academic staff.

With an API key

- The assistant can call DeepSeek or an OpenAI-compatible chat completion endpoint, depending on the selected provider.
- The prompt still includes local context and excludes sensitive fields such as passwords, tokens, hashes, credentials, and API keys.
- Conversation history is scoped to the logged-in user and retained according to the implemented retention logic.

Limitations

- Assistant answers are support information only, not official academic decisions.
- The assistant does not update any formal student ,teacher records or courses relevant records.
- External model calls require a valid key and network/service availability.

7. Troubleshooting for Users

Area	Problem	What to check	Suggested action
Login	Account cannot be found	The username may not exist in the users collection or may not have been created through Admin Management.	Confirm that the account exists in users or create it through Admin Management.
Login	Password is rejected	The stored password_hash may not match the intended password.	Check the imported account data or reset/update the account password.
Login	Only the built-in administrator can log in during local preview	Local H5 fallback mode may only contain the built-in administrator account unless more fallback/import data exists.	Use the provided administrator account for local fallback, or connect UniCloud and import the required users.
Login	Wrong dashboard opens after login	The account role controls routing. Supported app roles are student, teacher, and admin. academic_staff maps to admin; counselor maps to teacher.	Check and correct the user role data, then log in again.
Page access	Page redirects to login	The local session may be missing, invalid, or cleared.	Log in again. If it continues, clear browser storage or open a fresh browser session.
Page access	User cannot open a specific page	The page may not support the current role.	Confirm that the page is available for the user role. For example, Admin Management is for administrators, and student leave submission is for students.
Data loading	Dashboard, materials, evaluations, or leave records do not load	Read data may be cached briefly or the page may need a refresh after recent changes.	Refresh the page first, then retry the operation.
Data loading	Cloud-backed data is missing	The UniCloud service space may not be linked, cloud functions may not be uploaded, or database collections may not be imported.	Confirm the service space, upload required cloud functions, and import the related database collections.
Data loading	Data appears only as demo/fallback data	H5 preview without UniCloud may show local fallback data for some read-only views.	Use UniCloud setup for persistent database-backed workflows.
Leave workflow	Student cannot submit a leave request	The student may not be enrolled in the selected course, the date may not match a scheduled class, or the request may be submitted after class start.	Check enrollment, scheduled class date, and submission time before submitting again.
Leave workflow	Leave workflow has multi-teacher course issues	For courses with multiple teachers, the student may need to choose the relevant teacher before submitting related workflow records.	Ask the student to select the relevant teacher for that course, then retry.

Area	Problem	What to check	Suggested action
Leave workflow	Approved leave does not update attendance	The review-leave cloud function may not be uploaded, or attendance_records may be missing.	Confirm review-leave is uploaded and attendance_records exists. Approved leave should set the related attendance status to on_leave.
Leave workflow	Rejected leave changes attendance unexpectedly	Rejected leave requests should not change attendance.	Check the leave review result and related attendance record. Rejection should leave attendance unchanged.
Course evaluation	Course does not appear for evaluation	The course may not have ended in the current data, or the student may not be connected to the course offering.	Check course end date and student-course relationship.
Course evaluation	Evaluation for a multi-teacher course is blocked or grouped incorrectly	The relevant teacher may not have been selected for the course.	Choose the relevant teacher so feedback is counted under the correct teaching group.
Course evaluation	Evaluation summaries look incomplete	No evaluations may have been submitted, or get-evaluation-summary may be unavailable.	Submit evaluation data first and confirm the get-evaluation-summary cloud function is uploaded.
Course evaluation	User expects to see student identities	Summary views show aggregate scores and anonymous comments, not individual student identities.	Use the summary as anonymous feedback only.
Course materials	Students cannot see a material	The material may not be visible to students, or its available date/time may not have been reached.	Check visibility settings and the available date/time.
Course materials	Teachers/admins see materials that students cannot see	Teachers and administrators may see private or draft material records.	Mark the material as visible to students when it should appear in student views.
Course materials	File download does not work	The material may not have a valid file URL or cloud storage file ID, or the runtime may not support access.	Confirm the file URL/file ID and current runtime storage access.
AI assistant	Assistant returns a local fallback answer	No API key may be configured in AI Assistant settings.	Configure a valid provider, model, and API key if external model access is needed.
AI assistant	Assistant reports an AI service error	Provider, model name, API key, network access, or the ask-assistant cloud function may be unavailable.	Check provider settings, model name, API key, network/cloud availability, and the ask-assistant function.
AI assistant	Assistant does not change records	The assistant is for guidance and lookup only.	Use the normal workflow pages to submit leave, change attendance, edit materials, or update official records.
Course management	Course cannot be published	Required fields may be incomplete.	Check course code, course name, credits,

Area	Problem	What to check	Suggested action
			capacity, target major, cohort year, start/end date, total sessions, schedule slots, classrooms, and assigned teachers.
Course management	Course code already exists	Another course already uses the same course code.	Use a different course code or edit the existing course.
Course management	Course offering already exists	The same course, semester, and section number may already have been published.	Check existing offerings and change the section number if this should be a separate offering.
Course management	Date range cannot generate enough sessions	The selected date range and weekly schedule slots cannot produce the required total sessions.	Extend the date range, reduce total sessions, or add more weekly schedule slots.
Course management	Schedule slot is invalid	A slot may be missing weekday, start time, end time, or classroom, or the end time may not be later than the start time.	Complete each schedule slot and make sure each end time is later than its start time.
Course management	Schedule slots overlap inside the same course	Two schedule slots in the same course may overlap.	Adjust weekday or time ranges so the course's own slots do not overlap.
Course management	Major/cohort timetable conflict	The same major and cohort already has a class at that time.	Choose a different time slot or adjust the conflicting course schedule.
Course management	Classroom conflict	The selected classroom is already used by another scheduled session.	Choose another classroom or change the class time.
Course management	Invalid teacher, major, classroom, semester, or training plan	Related records may be missing, inactive, or inconsistent. Training plan must match the selected major and cohort year.	Refresh Admin Management and confirm the related records exist and match the selected course setup.
Course management	Course capacity is rejected	Capacity may not be positive, or it may be lower than current selected/enrolled students when editing.	Enter a positive capacity. When editing, keep capacity at or above the current selected student count.
Course management	Published course does not appear in timetable	Timetable filters may not match the course's major and cohort year, or cached data may need refresh.	Select the same major and cohort year used during publishing, then refresh the page.
Course management	Teacher does not appear in the teacher list	The teacher account or teacher profile may not exist or may not be active.	Confirm the teacher has a valid account and teacher profile.
Course management	Teacher time conflict is not detected	The current implementation validates selected teacher records but does not validate teacher availability across different courses.	Manually check teacher schedules before publishing.

Area	Problem	What to check	Suggested action
Course management	Course deletion fails	No specific course offering may be selected, or multiple offerings may exist for the same course.	Select the exact course offering to delete.
Course management	Course deletion removes more data than expected	Deleting a course offering also removes related sessions, enrollments, attendance records, leave records, evaluations, materials, recommendations, alerts, and grades.	Use deletion carefully and confirm the selected offering is the intended one.
Profile-change review	Submitted profile change does not appear for review	The page may need refresh, or submit-profile-change/review-profile-change may not be uploaded.	Refresh the dashboard or management page and confirm both cloud functions are uploaded.
Profile-change review	Approved or rejected profile change does not behave as expected	Approved changes should update the related profile record; rejected changes should keep the existing profile unchanged.	Check the review status and the related student or teacher profile record.